

XUSHENG ZHU, Ph.D.

Marie Skłodowska-Curie Actions Postdoctoral Fellow, University College London
Editor, IEEE TCOM | AE, IEEE WCL/CL/OJ-COMS | Guest Editor, IEEE TAES Special Section
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EDITORIAL PROFILE

Dr. Xusheng Zhu is an early-career researcher in 6G wireless communications and a Marie Skłodowska-Curie Actions Postdoctoral Fellow at University College London. His research focuses on reconfigurable wireless systems for 6G, with emphasis on fluid and movable antenna systems, reconfigurable/fluid intelligent surfaces, spatial and index modulation, transceiver design, mmWave MIMO, finite-blocklength and UAV communications, and AI-assisted wireless systems. He serves as an Editor of the *IEEE Transactions on Communications*, an Associate Editor of *IEEE Wireless Communications Letters*, an Associate Editor of *IEEE Communications Letters*, and an Associate Editor of the *IEEE Open Journal of the Communications Society*. He also serves as a Guest Editor for the *IEEE Transactions on Aerospace and Electronic Systems* Special Section on “Next-Generation Reconfigurable Antenna Systems for Aerospace Communications and Sensing.” His conference leadership includes Symposium Co-Chair of AIPIP 2026, Workshop General Co-Chair roles at IEEE GLOBECOM 2026, IEEE/CIC ICC 2026, and IEEE VTC 2026-Spring, as well as tutorial presentation and session-chair service at IEEE/CIC ICC 2026.

AREAS OF EDITORIAL EXPERTISE

- **6G physical-layer technologies:** fluid antenna systems, movable antennas, fluid/reconfigurable intelligent surfaces, programmable wireless environments, and intelligent radio environments.
- **Modulation, transceiver design, and optimization:** spatial modulation, space shift keying, spatial scattering modulation, index modulation, transceiver design, beamforming, precoding, and optimization.
- **Wireless system analysis:** communication theory and systems, mmWave MIMO, channel measurement, finite-blocklength and short-packet communications, UAV communications, RIS-aided networks, full-duplex communications, ISAC-oriented wireless systems, physical-layer security, and performance analysis.

SUMMARY OF ACADEMIC AND EDITORIAL RECORD

- **Journal publications:** 35 journal articles, including 19 first-/corresponding-author articles and 16 co-authored articles.
- **IEEE journal record:** Publications in *IEEE JSAC*, *IEEE TWC*, *IEEE TCOM*, *IEEE IoT Journal*, *IEEE TVT*, *IEEE TGCN*, *IEEE WCL*, *IEEE CL*, and *IEEE OJ-COMS*.
- **Conference publications:** 12 conference papers in IEEE ICC, IEEE GLOBECOM, IEEE VTC, IEEE/CIC ICC, IEEE GLOBECOM Workshops, and ICEIC.
- **Editorial service:** Editor/Associate Editor of four IEEE communications journals and Guest Editor of the *IEEE Transactions on Aerospace and Electronic Systems* Special Section on “Next-Generation Reconfigurable Antenna Systems for Aerospace Communications and Sensing.”
- **Reviewing service:** Regular reviewer for leading IEEE journals in wireless communications, communication theory and systems, vehicular technology, IoT, green communications, open-access communications, and communications letters.
- **Conference leadership:** Symposium Co-Chair for AIPIP 2026; Workshop General Co-Chair for IEEE GLOBECOM 2026 and IEEE VTC 2026-Spring; Workshop General Co-Chair and Track Chair for IEEE/CIC ICC 2026; Session Chair for the IEEE/CIC ICC 2026 FAS Workshop and IEEE ICC 2026 Workshops; Tutorial Presenter at IEEE/CIC ICC 2026.
- **Research leadership:** Principal Investigator of a Horizon Europe MSCA Postdoctoral Fellowship and an NSFC Youth Student Basic Research Project (Doctoral Student Program).
- **Broader technical contributions:** One standardization proposal, four patent applications/publications, and three co-authored 6G white papers.

ACADEMIC EXPERIENCE

University College London	London, UK
Marie Skłodowska-Curie Actions Postdoctoral Fellow	Apr. 2026–Mar. 2028
University College London	London, UK
Research Fellow	Jul. 2025–Mar. 2026
University College London	London, UK
Research Assistant	Apr. 2025–Jun. 2025
Hisense Group	Qingdao, China
Network Engineer, IT Department	Jul. 2017–Mar. 2019

EDUCATION

Shanghai Jiao Tong University	Shanghai, China
Ph.D. in Information and Communication Engineering	Sep. 2022–Jun. 2025
Lanzhou University	Lanzhou, China
M.E. in Electronics and Communication Engineering	Sep. 2019–Jun. 2022
Jilin University	Changchun, China
B.E. in Optical Communication	Sep. 2013–Jun. 2017

RESEARCH GRANTS & FELLOWSHIPS

- **Horizon Europe Marie Skłodowska-Curie Actions Postdoctoral Fellowship** 2026–2028
Role: Principal Investigator | Budget: EUR 276,000
Project No. 101269517: Spatial Multiplexing Fluid Antenna Systems (SM-FAS).
- **National Natural Science Foundation of China (NSFC) Youth Student Basic Research Project (Doctoral Student Program)** 2025–2026
Role: Principal Investigator | Budget: CNY 300,000

Project No. 624B2094: Key Technologies of Second-Order Spatial Modulation Based on Multi-Antenna and Reconfigurable Intelligent Surfaces.

• **SJTU Outstanding Ph.D. Graduate Development Scholarship**

2025–2026

Role: Principal Investigator | Budget: CNY 300,000

Research development scholarship for outstanding doctoral graduates.

EDITORIAL & PROFESSIONAL SERVICE

Journal Editorial Service

- **Editor**, *IEEE Transactions on Communications* (IEEE TCOM), Area: Communication Theory & Systems II, 2026–present.
- **Associate Editor**, *IEEE Wireless Communications Letters* (IEEE WCL), 2026–present.
- **Associate Editor**, *IEEE Communications Letters* (IEEE CL), Area: Wireless Communications, 2026–present.
- **Associate Editor**, *IEEE Open Journal of the Communications Society* (IEEE OJ-COMS), Area: Wireless Communications, 2026–present.
- **Guest Editor**, *IEEE Transactions on Aerospace and Electronic Systems* (IEEE TAES), Special Section on “Next-Generation Reconfigurable Antenna Systems for Aerospace Communications and Sensing,” 2026–2027.

Journal Reviewing Service

- **Regular reviewer for leading IEEE journals**, including *IEEE Journal on Selected Areas in Communications*, *IEEE Communications Magazine*, *IEEE Transactions on Wireless Communications*, *IEEE Transactions on Communications*, *IEEE Transactions on Vehicular Technology*, *IEEE Internet of Things Journal*, *IEEE Transactions on Green Communications and Networking*, *IEEE Systems Journal*, *IEEE Open Journal of the Communications Society*, *IEEE Open Journal of Vehicular Technology*, *IEEE Wireless Communications Letters*, and *IEEE Communications Letters*.
- **Reviewing expertise**: communication theory and systems, 6G physical-layer technologies, RIS/FRIS, fluid and movable antennas, MIMO, mmWave communications, spatial/index modulation, transceiver design, channel measurement, beamforming optimization, UAV communications, short-packet communications, ISAC, physical-layer security, and performance analysis.

Conference Leadership and Organization

- **Symposium Co-Chair**, AIPIP 2026, Symposium on “Intelligent Antennas, Reconfigurable Electromagnetic Technologies, and High-Frequency Links,” Hangzhou, China.
- **Workshops General Co-Chair**, IEEE GLOBECOM 2026, Macau, China; Workshop on Fluid Antenna System (FAS) for 6G.
- **Workshops General Co-Chair & Track Chair**, IEEE/CIC ICC 2026, Wuhan, China; Workshop on Fluid Antenna System (FAS) for 6G.
- **Workshops General Co-Chair**, IEEE VTC 2026-Spring, Nice, France; Workshop on Fluid Antenna System (FAS) for 6G.
- **Session Chair**, IEEE/CIC ICC 2026 Workshop on Fluid Antenna System (FAS) for 6G, Wuhan, China.
- **Session Chair**, IEEE ICC 2026 Workshops, WS-09: Fluid Antenna Systems for 6G, Glasgow, UK.

Tutorial Presentation

- **Tutorial Presenter**, IEEE/CIC ICC 2026, Wuhan, China.

Technical Program Committee Member

- **2026**: IEEE GLOBECOM Workshops (FAS for 6G), IEEE ICC Workshops (FAS for 6G), IEEE GLOBECOM SAC-RISSE, IEEE ICC SPC, IEEE ICC WC, and IEEE WCNC Workshops.
- **2025**: IEEE GLOBECOM SAC-RIS, IEEE VTC 2025-Spring, IEEE ICNC 2025, and ICRAMET 2025.
- **2023–2024**: IEEE GLOBECOM 2024, IEEE VTC 2024-Fall, and IEEE VTC 2023-Fall.

HONORS & AWARDS

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| • Shanghai Outstanding Graduate, Shanghai Jiao Tong University, Top 1%, provincial level | 2025 |
| • Shangjun Scholarship, Shanghai Jiao Tong University, Top 3%, university level | 2025 |
| • Excellent Teaching Assistant Nomination Award, Shanghai Jiao Tong University | 2025 |
| • National Scholarship, Shanghai Jiao Tong University, Top 1%, national level | 2024 |
| • Second Prize, Excellent Annual Conference Paper, Shanghai Institute of Communications | 2024 |
| • Merit Student, Shanghai Jiao Tong University, Top 10% | 2024 |
| • Outstanding Graduate, Lanzhou University, university level | 2022 |
| • Outstanding Graduate Student, Lanzhou University, university level | 2022 |
| • Second Prize, Electronic Design Contest, Jilin University | 2016 |
| • Third Prize, Electronic Design Contest, Jilin University | 2014 |

FULL PUBLICATION RECORD

First-Author and Corresponding-Author Journal Articles

*Corresponding author.

1. Z. Jiang, Y. Zhou, **X. Zhu***, W. Chen, and K.-K. Wong, “Joint optimization for multi-user transmissive RIS-MIMO systems,” *IEEE Transactions on Vehicular Technology*, doi: 10.1109/TVT.2026.3694678, 2026.
2. **X. Zhu**, K.-K. Wong, H. Xu, H. Xiao, H. Hong, H. Shin, and Y. Zhang, “Fluid antenna systems: A geometric approach to error probability and fundamental limits,” *IEEE Transactions on Wireless Communications*, vol. 25, pp. 17195–17209, May 2026.
3. **X. Zhu**, K.-K. Wong, H. Hong, H. Xiao, H. Xu, T. Wu, and C.-B. Chae, “Fluid antenna system-enabled UAV communications in the finite blocklength regime,” *IEEE Transactions on Wireless Communications*, vol. 25, pp. 16714–16729, May 2026.
4. **X. Zhu**, K.-K. Wong, B. Tang, W. Chen, and C.-B. Chae, “Fluid reconfigurable intelligent surface enabling secure wireless communications,” *IEEE Wireless Communications Letters*, vol. 15, pp. 2408–2412, Apr. 2026.
5. **X. Zhu**, Q. Wu, W. Chen, Y. Liu, M. Jian, and D. da Costa, “Spatial scattering shift keying for mmWave MIMO systems,” *IEEE Transactions on Communications*, vol. 73, no. 12, pp. 15337–15351, Dec. 2025.

6. **X. Zhu**, Q. Wu, W. Chen, Y. Zhou, Y. Hou, and R. Liu, "Hybrid IRS and relay network for enhanced wireless communication: Analysis and optimization," *China Communications*, vol. 22, no. 11, pp. 17–27, Nov. 2025.
7. **X. Zhu**, Q. Wu, W. Chen, X. He, Y. Hou, and Y. Zhang, "Discrete RIS enhanced space shift keying MIMO system via reflecting beamforming optimization," *IEEE Wireless Communications Letters*, vol. 14, no. 8, pp. 2471–2475, Aug. 2025.
8. **X. Zhu**, Q. Wu, W. Chen, Z. Zhang, X. Bai, and X. Guan, "Towards spatial scattering modulation: Detector design and error probability analysis," *IEEE Transactions on Wireless Communications*, vol. 24, no. 10, pp. 8520–8534, Oct. 2025.
9. **X. Zhu**, Q. Wu, and W. Chen, "Transmissive RIS transmitter enabled spatial modulation for MIMO systems," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 3, pp. 899–911, Mar. 2025.
10. **X. Zhu**, Q. Wu, and W. Chen, "On the performance of RIS-aided spatial modulation for downlink transmission," *IEEE Transactions on Wireless Communications*, vol. 23, no. 11, pp. 16203–16217, Nov. 2024.
11. **X. Zhu**, W. Chen, Q. Wu, W. Fang, C. Huang, and J. Li, "Robust analysis of full-duplex two-way space shift keying with RIS systems," *IEEE Transactions on Communications*, vol. 72, no. 11, pp. 7233–7249, Nov. 2024.
12. **X. Zhu**, W. Chen, Q. Wu, J. Li, N. Cheng, F. Chen, and C. Li, "Performance analysis of RIS-aided double spatial scattering modulation for mmWave MIMO systems," *IEEE Transactions on Wireless Communications*, vol. 23, no. 6, pp. 6139–6155, Jun. 2024.
13. **X. Zhu**, W. Chen, Q. Wu, Z. Li, J. Li, S. Zhang, and M. Ding, "Reconfigurable-intelligent-surface-aided space-shift keying with imperfect CSI," *IEEE Internet of Things Journal*, vol. 11, no. 7, pp. 11480–11495, Apr. 2024.
14. **X. Zhu**, W. Chen, Q. Wu, Z. Liu, and J. Li, "Performance of RIS-assisted full-duplex space shift keying with imperfect self-interference cancellation," *IEEE Wireless Communications Letters*, vol. 13, no. 1, pp. 128–132, Jan. 2024.
15. **X. Zhu**, W. Chen, Z. Li, Q. Wu, Z. Zhang, K. Wang, and J. Li, "RIS-aided spatial scattering modulation for mmWave MIMO transmissions," *IEEE Transactions on Communications*, vol. 71, no. 12, pp. 7378–7392, Dec. 2023.
16. **X. Zhu**, L. Yuan, Q. Li, L. Jin, X. Nie, C. Pan, and J. Zhang, "RIS-assisted full-duplex space shift keying: System scheme and performance analysis," *IEEE Transactions on Green Communications and Networking*, vol. 7, no. 4, pp. 1981–1995, Dec. 2023.
17. **X. Zhu**, W. Chen, Z. Li, Q. Wu, and J. Li, "Quadrature spatial scattering modulation for mmWave transmission," *IEEE Communications Letters*, vol. 27, no. 5, pp. 1462–1466, May 2023.
18. **X. Zhu**, L. Yuan, Q. Li, L. Jin, X. Nie, C. Pan, and J. Zhang, "On the performance of 3-D spatial modulation over measured indoor channels," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 2, pp. 2110–2115, Feb. 2022.
19. **X. Zhu**, L. Yuan, K. J. Kim, Q. Li, and J. Zhang, "Reconfigurable intelligent surface-assisted spatial scattering modulation," *IEEE Communications Letters*, vol. 26, no. 1, pp. 192–196, Jan. 2022.

Other Co-Authored Journal Publications

1. H. Hong, K.-K. Wong, N. Waqar, **X. Zhu**, C. Rao, W. Zhang, F. R. Ghadi, R. Murch, and C.-B. Chae, "Fluid antenna multiple access for 6G: A holistic review," *IEEE Open Journal of the Communications Society*, vol. 7, pp. 2607–2633, Mar. 2026.
2. C. Rao, K.-K. Wong, S. Xu, **X. Zhu**, Y. Zhang, and C.-B. Chae, "Codebook-based port selection and combining for CSI-free uplink fluid antenna multiple access," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 13401–13415, Mar. 2026.
3. H. Xiao, X. Hu, K.-K. Wong, **X. Zhu**, H. Hong, and C.-B. Chae, "Fluid reconfigurable intelligent surface with element-level pattern reconfigurability: Beamforming and pattern co-design," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 10791–10806, Jan. 2026.
4. C. Huang, W. Chen, Q. Wu, **X. Zhu**, Z. Li, Y. Wang, and J. Yuan, "Dual-IRS aided near/hybrid-field SWIPT: Passive beamforming and independent antenna power splitting design," *IEEE Transactions on Communications*, vol. 74, pp. 2841–2856, Jan. 2026.
5. X. Nie, X. Wan, S. Li, Q. Zhou, L. Liao, **X. Zhu**, and J. Zhang, "Heterogeneous multi-agent reinforcement learning for resource allocation in NOMA-enabled mmWave IAB networks," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 10816–10828, Dec. 2025.
6. Z. Li, W. Chen, H. Qin, Q. Wu, **X. Zhu**, Z. Zhang, and J. Li, "Towards TMA-based transmissive RIS transceiver enabled downlink communication networks: A consensus-ADMM approach," *IEEE Transactions on Communications*, vol. 73, no. 4, pp. 2832–2846, Apr. 2025.
7. Z. Li, W. Chen, **X. Zhu**, Q. Wu, G. Ni, and S. Zhang, "Transmissive RIS transceiver enabled multi-stream communication systems: Design, optimization and analysis," *IEEE Internet of Things Journal*, vol. 12, no. 5, pp. 5985–6000, Mar. 2025.
8. Z. Zhang, W. Chen, Q. Wu, Z. Li, **X. Zhu**, J. Chen, and N. Cheng, "Multiple intelligent reflecting surfaces collaborative wireless localization system," *IEEE Transactions on Wireless Communications*, vol. 24, no. 1, pp. 134–148, Jan. 2025.
9. X. Nie, C. Wang, T. Zhou, Q. Zhou, **X. Zhu**, and J. Zhang, "Mobility-aware cooperative caching in IoVs based on secure asynchronous federated and deep reinforcement learning," *IEEE Internet of Things Journal*, vol. 12, no. 12, pp. 20572–20588, Jun. 2025.
10. Y. Zhou, W. Chen, Q. Wu, **X. Zhu**, and N. Cheng, "Movable antenna empowered downlink NOMA systems: Power allocation and antenna position optimization," *IEEE Wireless Communications Letters*, vol. 13, no. 10, pp. 2772–2776, Oct. 2024.
11. X. He, W. Chen, Q. Wu, **X. Zhu**, and N. Cheng, "Movable antenna enhanced NOMA short-packet transmission," *IEEE Communications Letters*, vol. 28, no. 9, pp. 2196–2200, Sep. 2024.
12. Z. Zhang, W. Chen, Q. Wu, Z. Li, **X. Zhu**, and J. Yuan, "Intelligent omni surfaces assisted integrated multi-target sensing and multi-user MIMO communications," *IEEE Transactions on Communications*, vol. 72, no. 8, pp. 4591–4606, Aug. 2024.
13. Z. Li, W. Chen, Q. Wu, **X. Zhu**, H. Qin, K. Wang, and J. Li, "Toward transmissive RIS transceiver enabled uplink communication systems: Design and optimization," *IEEE Internet of Things Journal*, vol. 11, no. 4, pp. 6788–6801, Feb. 2024.
14. Z. Liu, W. Chen, Q. Wu, Z. Li, **X. Zhu**, Q. Wu, and N. Cheng, "Enhancing robustness and security in ISAC network design: Leveraging transmissive reconfigurable intelligent surface with RSMA," *IEEE Transactions on Communications*, vol. 73, no. 10, pp. 9581–9596, Oct. 2025.
15. W. Fang, W. Chen, Q. Wu, **X. Zhu**, Q. Wu, and N. Cheng, "Channel characterization of IRS-assisted resonant beam communication systems," *IEEE Transactions on Communications*, vol. 73, no. 9, pp. 7381–7397, Sept. 2025.
16. S. Du, S. Chen, S. Cheng, J. He, D. Zhao, **X. Zhu**, L. Lian, X. Tu, Q. Zhao, and Y. Zhang, "Dust storm detection for ground-based stations with imbalanced machine learning," *Environmental Modelling and Software*, Mar. 2025.

Conference Proceedings

1. **X. Zhu**, F. R. Ghadi, T. Wu, K. Meng, C. Wang, and G. Zhou, "On the fundamental scaling laws of fluid antenna systems," *IEEE Globecom Workshops (GC Wkshps)*, Taipei, Taiwan, 2025, pp. 436-441.
2. **X. Zhu**, K.-K. Wong, H. Hong, H. Xu, and L. Wang, "Fundamental limits of FAS-enabled URLLC: Finite blocklength coding and port switching delay," *Global 6G Conference (Global 6G)*, Nanjing, China, 2026, pp. 1-6.
3. **X. Zhu**, K.-K. Wong, H. Hong, H. Xiao, H. Xu, T. Wu, and C.-B. Chae, "UAV-enabled short-packet communication via fluid antenna systems," in *Proc. IEEE ICC*, Glasgow, UK, 2026.
4. **X. Zhu**, F. R. Chadi, T. Wu, K. Meng, C. Wang, and G. Zhou, "On the fundamental scaling laws of fluid antenna systems," in *Proc. IEEE GLOBECOM Workshops*, Taipei, 2025.
5. **X. Zhu**, Q. Wu, W. Chen, and Y. Zhou, "Rethinking the detectors design of spatial scattering modulation for mmWave MIMO systems," in *Proc. IEEE 101st VTC2025-Spring*, Oslo, Norway, 2025.
6. **X. Zhu**, W. Chen, and Q. Wu, "Performance evaluation of RIS-assisted spatial modulation for downlink transmission," in *Proc. IEEE ICC Workshops*, Denver, USA, 2024.
7. **X. Zhu**, W. Chen, Z. Li, Q. Wu, Z. Zhang, K. Wang, and J. Li, "On the performance of RIS-aided spatial scattering modulation for mmWave MIMO transmission," in *Proc. IEEE GLOBECOM*, Kuala Lumpur, Malaysia, Dec. 2023.
8. **X. Zhu**, W. Chen, Q. Wu, and L. Wang, "Performance analysis of RIS-aided space shift keying with channel estimation errors," in *Proc. IEEE/CIC ICC*, Dalian, China, Aug. 2023.
9. H. Hong, H. Xu, K. -K. Wong, **X. Zhu**, and R. Liu, "OFDM-based CUMA for 5G NR networks," *2026 Global 6G Conference (Global 6G)*, Nanjing, China, 2026, pp. 1-6.
10. Y. Zhou, W. Chen, Q. Wu, and X. Zhu, "Movable antenna-enabled multi-user NOMA wireless systems," *IEEE Globecom Workshops (GC Wkshps)*, Taipei, Taiwan, 2025, pp. 508-513.
11. X. Nie, C. Zhang, C. Wang, Q. Zhou, **X. Zhu**, and J. Zhang, "Structure-aware GNN-transformer for content prediction and caching in vehicular networks," in *Proc. ICEIC*, Macau, China, 2026.
12. X. Nie, G. Sun, S. Li, C. Zhang, **X. Zhu**, and J. Zhang, "Topology-aware multi-hop task offloading via GNN-A routing in vehicular networks," in *Proc. ICEIC*, Macau, China, 2026.
13. Z. Jiang, **X. Zhu**, Q. Wu, and W. Chen, "Detector design and ABEP analysis for spatial scattering shift keying in mmWave systems," in *Proc. IEEE/CIC ICC*, Shanghai, China, Aug. 2025.
14. T. Hao, C. Shi, B. Xia, **X. Zhu**, Y. Guo, L. Ding, and F. Yang, "Mutual coupling exploitation for ISAC system with tunable antenna load," in *Proc. IEEE 101st VTC-Spring*, Jun. 2025.
15. H. Hong, H. Xu, and **X. Zhu**, "Geometric shaping non-uniform constellation design in fluid antenna system," in *Proc. IEEE/CIC ICC*, Shanghai, China, Aug. 2025.
16. Z. Zhang, W. Chen, Q. Wu, Z. Li, **X. Zhu**, J. Chen, and N. Cheng, "Collaborative localization with multiple intelligent reflecting surfaces," in *Proc. IEEE GLOBECOM Workshops*, Dec. 2024.

PATENTS, STANDARDIZATION & WHITE PAPERS

Standardization Proposal

- **X. Zhu**, Q. Wu, and W. Chen, "Reflecting phase shift design for discrete RIS assisted SSK MIMO systems," *IMT-2030 (6G) RIS Task Group*, Proposal No. IMT-2030_WX_XX_RIS_202410014, 2024.

Patents

- Q. Wu, **X. Zhu**, and W. Chen, "A mmWave spatial scattering shift keying modulation system and detection method," Patent Application, China.
- Q. Wu, **X. Zhu**, and W. Chen, "A hybrid intelligent reflecting surface and decode-and-forward relay wireless communication system and optimization method," Patent Application, China.
- Z. Li, W. Chen, Z. Zhang, Z. Liu, and **X. Zhu**, "Transmissive RIS transceiver uplink multi-user communication system and method," Published.
- Z. Li, W. Chen, Z. Zhang, Z. Liu, **X. Zhu**, and G. Ni, "Transmissive RIS transceiver downlink multi-stream communication system and method," Published.

Co-Author 6G White Papers

- "Research progress and network deployment challenges of RIS," *IMT-2030 (6G) Promotion Group*, Apr. 2024.
- "6G Near-field Technologies White Paper," Apr. 2024.
- "6G Near-field Technologies White Paper 2.0," Apr. 2025.